

$$\begin{aligned}
& \delta_{112} \delta_{134} \rightarrow -\frac{1}{2} c_Y^2 \overline{y}_d^{i2i3} y_u^{i1i4} + \\
& \frac{1}{16\pi^2} \left(\frac{1}{72} \frac{1}{m_b^2} \left(c_Y^2 \left(9 \overline{y}_d^{i2i3} y_u^{pi4} \left(\overline{y}_d^{pr} y_d^{i1r} (-1 + s_Y^2) - 3 \overline{y}_u^{pr} y_u^{i1r} (1 + c_Y^2) \right) - \right. \right. \right. \\
& \quad 9 y_u^{i1i4} \left(-\overline{y}_d^{i2p} y_d^{rp} \overline{y}_u^{ri3} (-1 + s_Y^2) + 3 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{pr} (1 + c_Y^2) + \right. \\
& \quad \left. \left. 2 \overline{y}_u^{i2i3} (5 g_1^2 + 3 (g_2^2 + 4 g_3^2 - 4 s_Y^2 \overline{y}_u^{pr} y_u^{pr})) \right) \right) - 16 g_3^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} \right) - \\
& \quad 8 g_3^2 (s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} + c_Y^2 \overline{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4} + \frac{2}{3} g_3^4 LF_{3,0}[m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad g_3^4 LF_{4,-1}[m_3] \delta_{i1i2} \delta_{i3i4} - \frac{16}{15} g_3^4 LF_{5,-2}[m_3] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{2} \sum p_C y_g g_1^2 \frac{1}{m_b^2} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 y_C y - 2 s_2 y_C s_Y) LF_{1,0}[m_d^p] + \\
& \quad \frac{2}{9} \sum p_C g_3^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum p_C g_3^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{8}{45} \sum p_C g_3^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\
& \quad 3 s_Y c_Y \frac{1}{m_b^4} \overline{y}_d^{pr} y_d^{pr} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 y_C + s_Y c_Y) LF_{1,0}[m_d^r] - \\
& \quad \frac{1}{2} \sum p_C y_g g_1^2 \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 y_C y - 2 s_2 y_C s_Y) LF_{1,0}[m_e^p] - \\
& \quad s_Y c_Y \frac{1}{m_b^4} \overline{y}_e^{pr} y_e^{pr} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 y_C + s_Y c_Y) LF_{1,0}[m_e^r] + \\
& \quad \frac{1}{2} c_Y \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (-2 s_Y \overline{y}_e^{pr} y_e^{pr} (s_2 y_C + s_Y c_Y) + \sum p_C g_1^2 (c_2 y_C y - 2 s_2 y_C s_Y)) LF_{1,0}[m_l^p] - \\
& \quad \frac{1}{2} c_Y \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (6 s_Y \overline{y}_d^{pr} y_d^{pr} (s_2 y_C + s_Y c_Y) + 6 \overline{y}_u^{pr} y_u^{pr} (c_Y^3 - s_2 y_C s_Y) + \\
& \quad \sum p_C g_1^2 (c_2 y_C y - 2 s_2 y_C s_Y)) LF_{1,0}[m_q^p] + \frac{4}{9} \sum p_C g_3^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{5}{6} \sum p_C g_3^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{45} \sum p_C g_3^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \sum p_C y_g g_1^2 \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 y_C y - 2 s_2 y_C s_Y) LF_{1,0}[m_u^p] + \frac{2}{9} \sum p_C g_3^4 LF_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\
& \quad - \frac{5}{12} \sum p_C g_3^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{8}{45} \sum p_C g_3^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\
& \quad 3 c_Y \frac{1}{m_b^4} \overline{y}_u^{pr} \overline{y}_u^{i2i3} y_u^{pr} y_u^{i1i4} (c_Y^3 - s_2 y_C s_Y) LF_{1,0}[m_u^r] + \\
& \quad \frac{1}{4} c_Y \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_Y (g_1^2 (1 - 3 c_2 y_C^2) - 3 g_2^2 (-1 + c_2 y_C^2)) + 3 s_4 y_C s_Y (g_1^2 + g_2^2)) \\
& \quad LF_{1,0}[m_b] + \frac{1}{4} \frac{1}{m_b^2} c_Y^2 (3 \overline{y}_u^{i2i3} y_u^{pi4} (-s_Y^2 \overline{y}_d^{pr} y_d^{i1r} + c_Y^2 \overline{y}_u^{pr} y_u^{i1r}) - \\
& \quad y_u^{i1i4} (2 \overline{y}_u^{i2i3} (g_1^2 + 3 g_2^2) + 3 s_Y^2 \overline{y}_d^{i2p} y_d^{rp} \overline{y}_u^{ri3} - 3 c_Y^2 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{pr})) LF_{1,1}[m_b] + \\
& \quad \frac{1}{12} (c_Y^2 (6 s_Y^2 \overline{y}_d^{pr} y_d^{i1r} \overline{y}_u^{i2i3} y_u^{pi4} + 3 y_u^{i1i4} (\overline{y}_u^{i2i3} (g_1^2 + 3 g_2^2) + 2 s_Y^2 \overline{y}_d^{i2p} y_d^{rp} \overline{y}_u^{ri3}) - \\
& \quad 8 g_3^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2}) - 4 g_3^2 (s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} + c_Y^2 \overline{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4}) LF_{1,2}[m_b] + \\
& \quad \frac{1}{36} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_q^{i1}] - \frac{1}{72} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_q^{i1}] + \\
& \quad - \frac{1}{216} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad - \frac{1}{108} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{1}{36} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_q^{i2}] - \\
& \quad - \frac{1}{72} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_q^{i2}] + \frac{1}{216} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{1}{216} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{1}{216} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{9} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_u^{i3}] - \\
& \quad - \frac{2}{9} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_u^{i3}] + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{2}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{2}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{9} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_1, m_u^{i4}] - \\
& \quad - \frac{2}{9} g_1^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{2,1,-1}[m_1, m_u^{i4}] + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{2}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad - \frac{2}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + g_1^2 \frac{1}{m_b^4} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,-1}[m_1, \tilde{\mu}] + \\
& \quad 2 m_1 s_Y \tilde{\mu} c_Y g_1^2 \frac{1}{m_b^4} \overline{y}_u^{i2i3} y_u^{i1i4} (-2 c_Y^2 + s_Y^2) LF_{1,1,0}[m_1, \tilde{\mu}] + \\
& \quad \frac{3}{4} g_2^2 \frac{1}{m_b^2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} LF_{1,1,0}[m_2, m_q^{i1}] - \\
& \quad \frac{3}{8} g_2$$